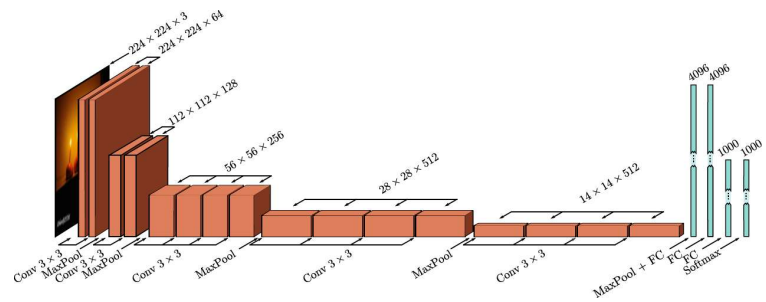
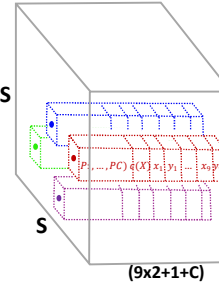
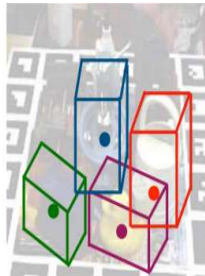
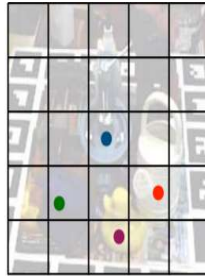


Agenda

- Convolutional Neural Network



$$N_{\text{out}} = \left\lfloor \frac{N_{\text{in}} + 2P - K}{S} \right\rfloor + 1$$



3

Networks for images

- Problems with fully-connected networks
 - Size
 - 224×224 RGB image = 150,528 dimensions
 - Hidden layers generally larger than inputs
 - One hidden layer = $150,520 \times 150,528$ weights = ~22 billion
 - Nearby pixels statistically related
 - Should be stable under transformations
 - Don't want to re-learn appearance at different parts of image

4

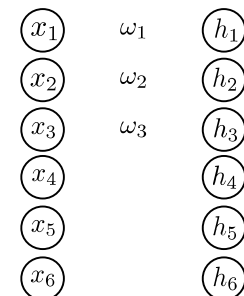
CNN Concepts

- Identical, Independent, parallel, output can match input shape, special case of a FCN
- Kernel, kernel width
 - Technical definition vs definition in practice (correlation vs convolution)
 - Valid convolution
- Stride, and dilation, pooling, down-sampling
- Channels, Receptive field
- 2D Convolution
- Upsampling

5

Channels

- The convolutional operation averages together the inputs
- Plus passes through ReLU function
 - Has to lose information
- Solution:
 - Apply several convolutions and stack them in channels
 - Called feature maps

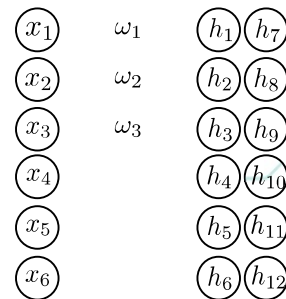


6

Two Output Channels, One Input Channel

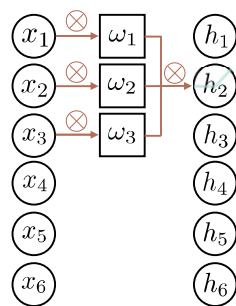
○ Solution:

- Apply several convolutions and stack them in channels
- **Called feature maps**



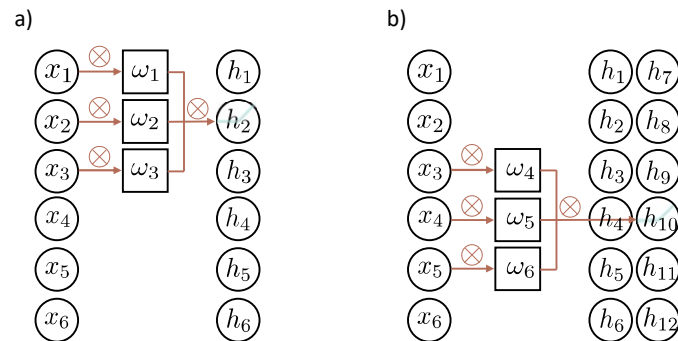
7

Two Output Channels, One Input Channel



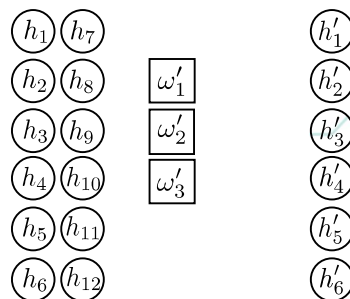
8

Two Output Channels, One Input Channel



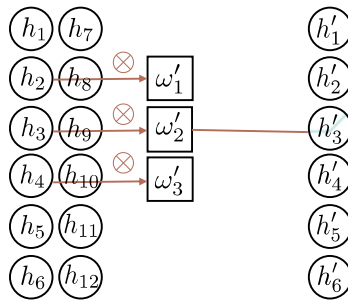
9

Two Input Channels, One Output Channel



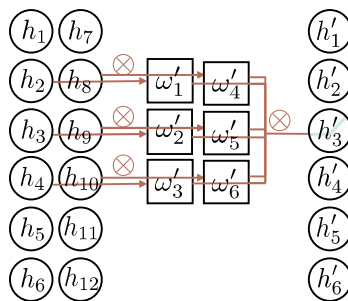
10

Two Input Channels, One Output Channel



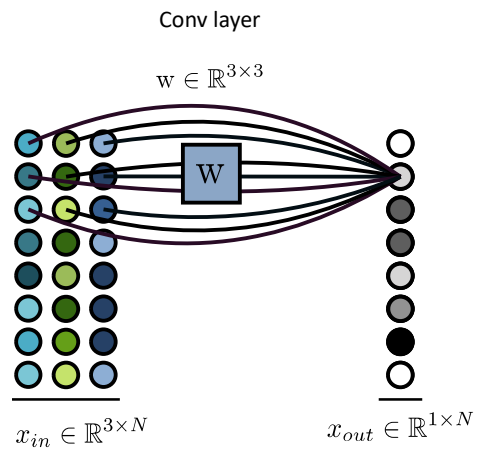
11

Two Input Channels, One Output Channel



12

Three input channel, one output channel



One dimensional example

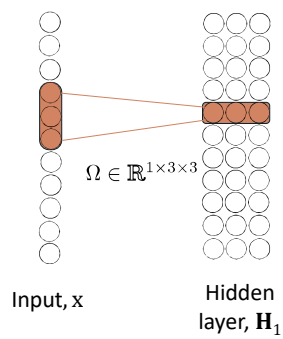
$$x_{out} = \sum_c w[c, :] \star x_{in}[c, :] + b[c]$$

13

Multiple Input & Output Channels

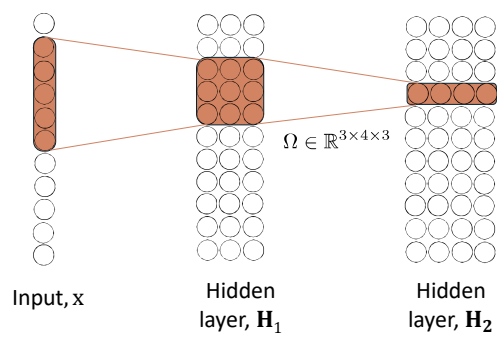
14

Receptive fields



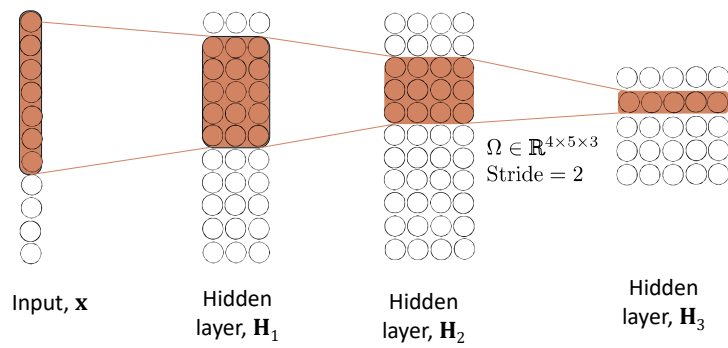
15

Receptive fields



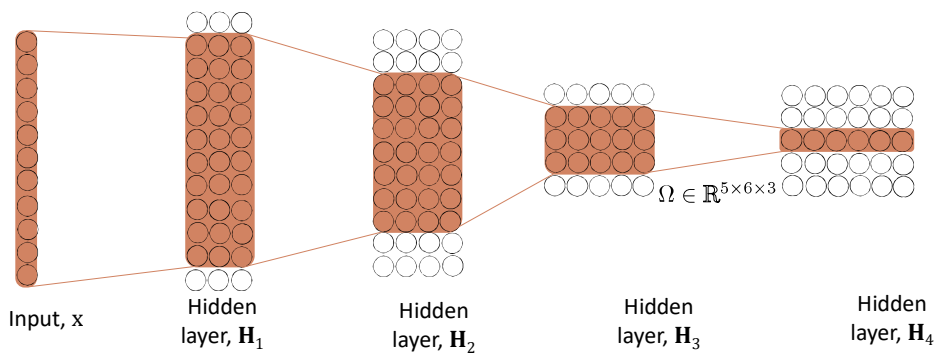
16

Receptive fields



17

Receptive Fields



18

2D Convolution

○ Convolution in 2D

- Weighted sum over a $K \times K$ region
- $K \times K$ weights

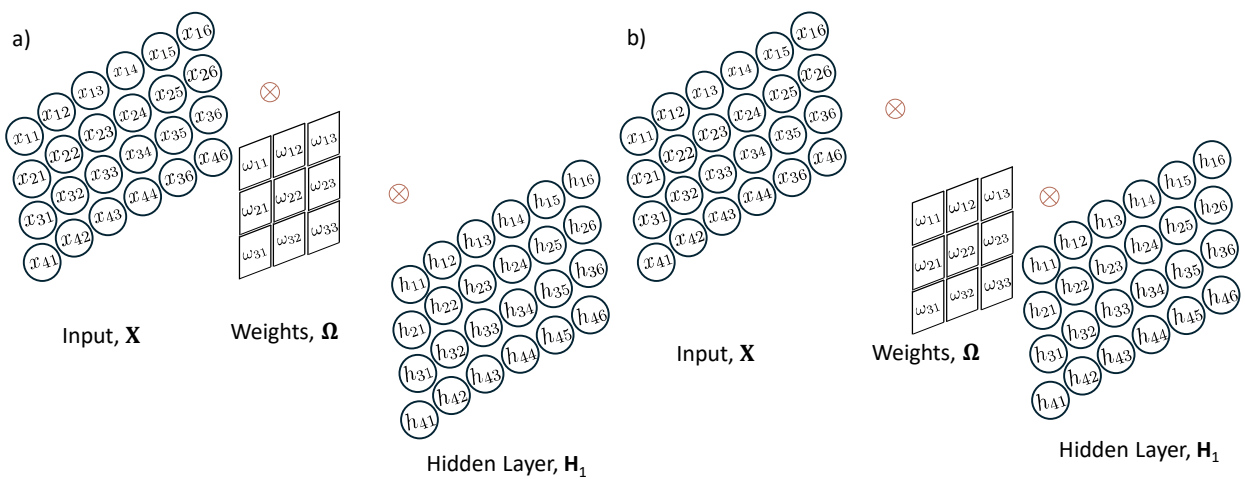
$$\sum_{m=1}^3 \sum_{n=1}^3 \omega_{m,n} x_{i+m-2, j+n-2}$$

- Build into a convolutional layer by adding bias and passing through activation function

$$h_{i,j} = a \left[\beta + \sum_{m=1}^3 \sum_{n=1}^3 \omega_{m,n} x_{i+m-2, j+n-2} \right]$$

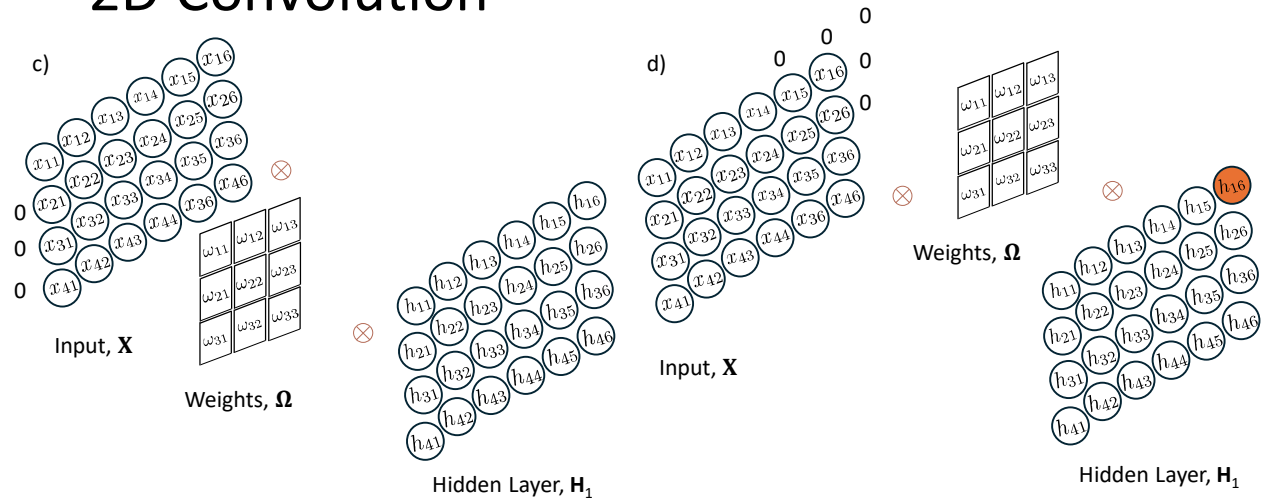
20

2D Convolution



21

2D Convolution



22

Toeplitz matrix

$$\begin{pmatrix} a & b & c & d & e \\ e & a & b & c & d \\ d & e & a & b & c \\ c & d & e & a & b \\ b & c & d & e & a \end{pmatrix}$$

$$y = T x$$

Each diagonal is the same
from top to bottom

$$y = T x$$

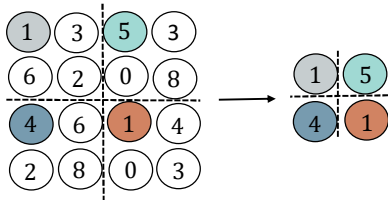
Example Block Toeplitz matrix (for 2D case)

$$\begin{bmatrix} e & f & 0 & 0 & h & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ d & e & f & 0 & g & h & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & d & e & f & 0 & g & h & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & d & e & 0 & 0 & g & h & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b & c & 0 & 0 & e & f & 0 & 0 & h & i & 0 & 0 & 0 & 0 & 0 \\ a & b & c & 0 & d & e & f & 0 & g & h & i & 0 & 0 & 0 & 0 \\ 0 & a & b & c & 0 & d & e & f & 0 & g & h & i & 0 & 0 & 0 \\ 0 & 0 & a & b & 0 & 0 & d & e & 0 & 0 & g & h & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & b & c & 0 & 0 & e & f & 0 & 0 & h & i & 0 \\ 0 & 0 & 0 & 0 & a & b & c & 0 & d & e & f & 0 & g & h & i \\ 0 & 0 & 0 & 0 & 0 & a & b & c & 0 & d & e & f & 0 & g & h \\ 0 & 0 & 0 & 0 & 0 & 0 & a & b & c & 0 & d & e & f & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & a & b & c & 0 & d & e & f & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a & b & c & 0 & d & e & f \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a & b & c & 0 & d & e \end{bmatrix}$$

23

Downsampling

a)

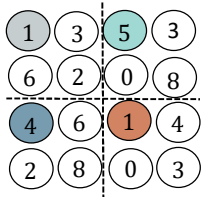


Sample every other
position (equivalent to
stride two)

24

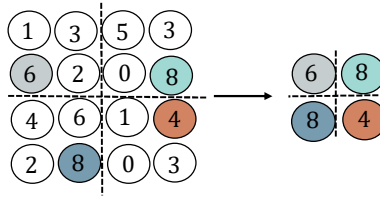
Downsampling

a)



Sample every other
position (equivalent to
stride two)

b)

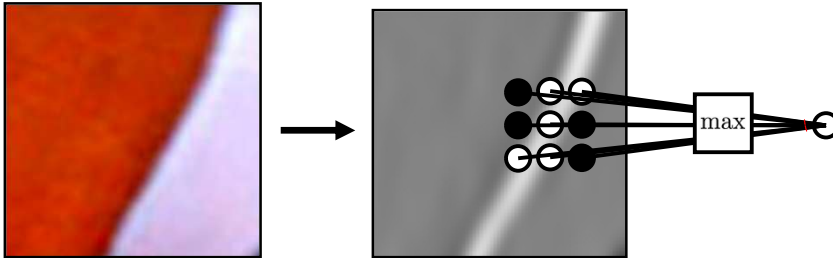


Max pooling
(partial invariance to
translation)

25

Pooling — Why?

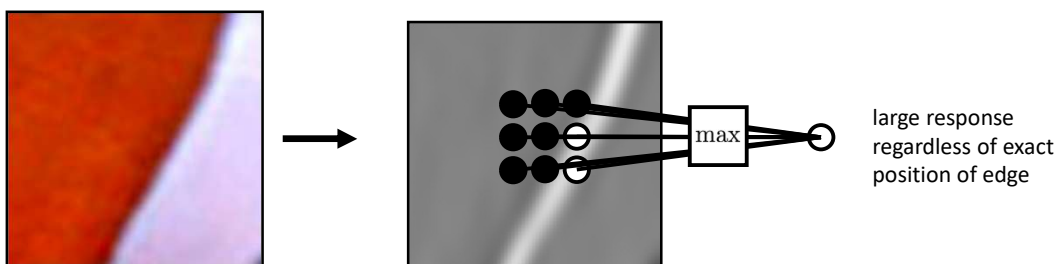
Pooling across spatial locations achieves stability w.r.t. small translations:



26

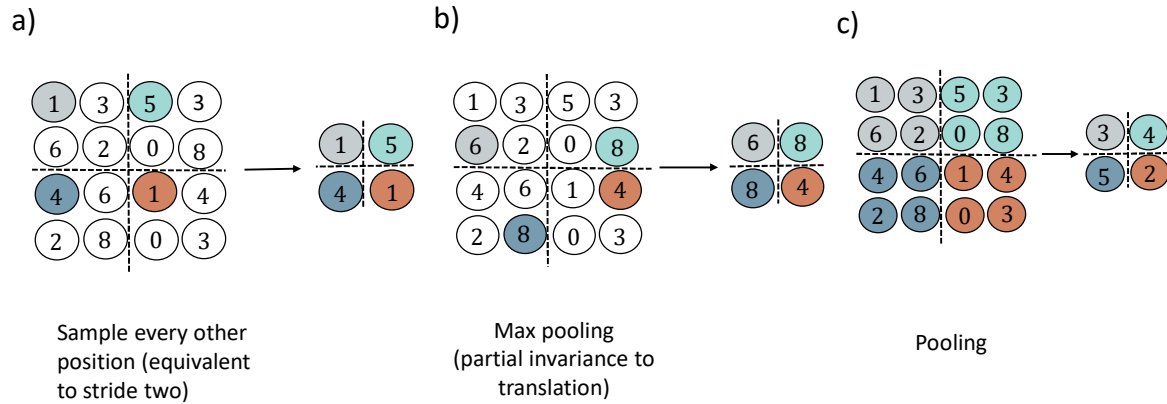
Pooling — Why?

Pooling across spatial locations achieves stability w.r.t. small translations:



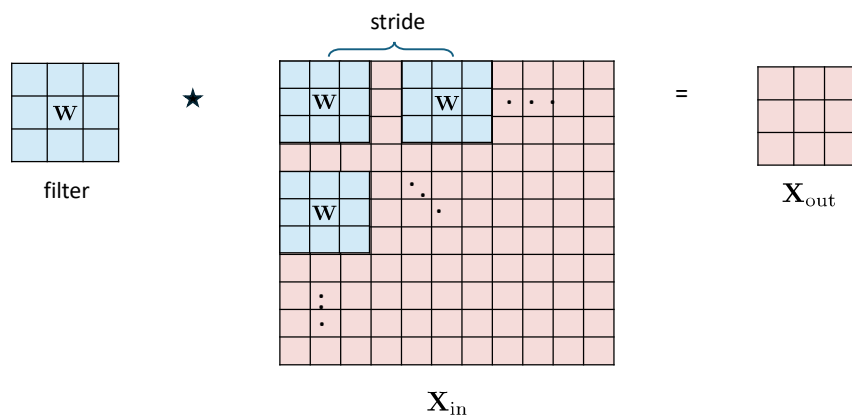
27

Downsampling



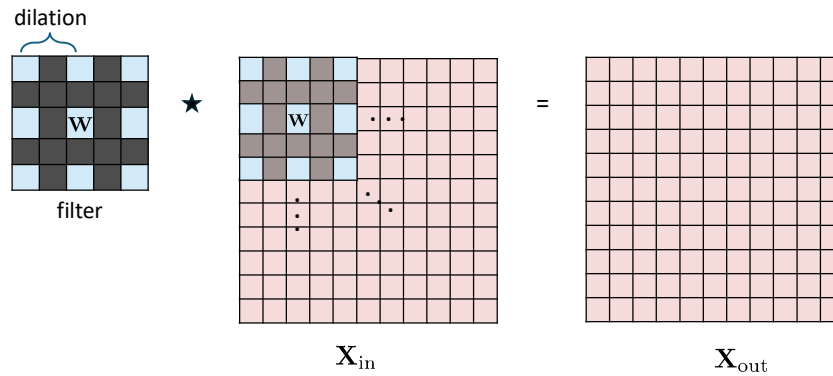
28

Strided operations (2D)



29

Dilated filter

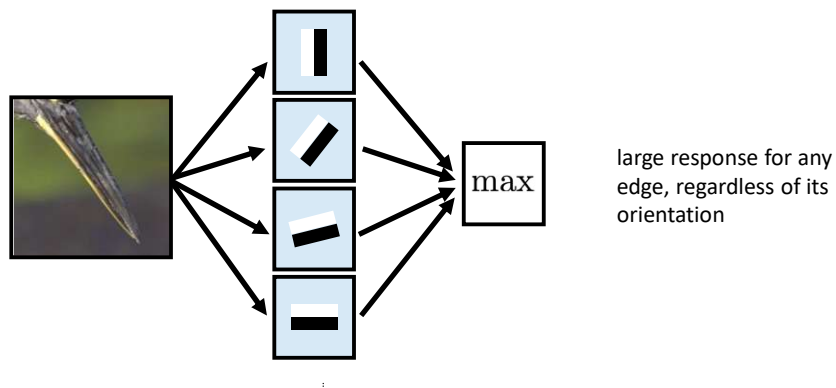


Covers a large receptive field with fewer parameters.

30

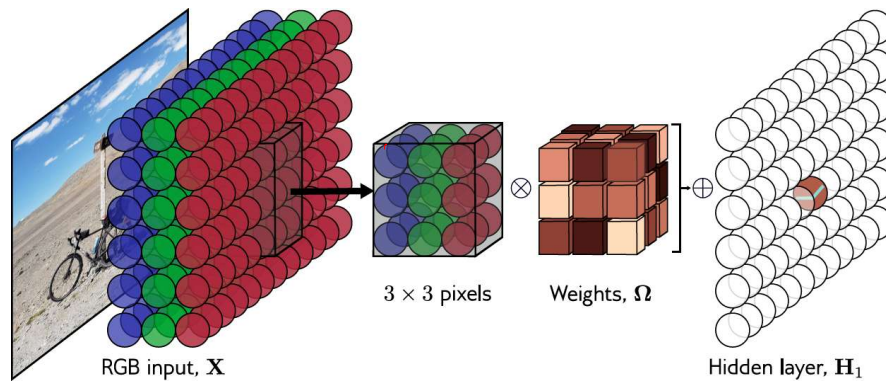
Pooling Across Channels — Why?

Pooling across feature channels (filter outputs) can achieve other kinds of invariances:



31

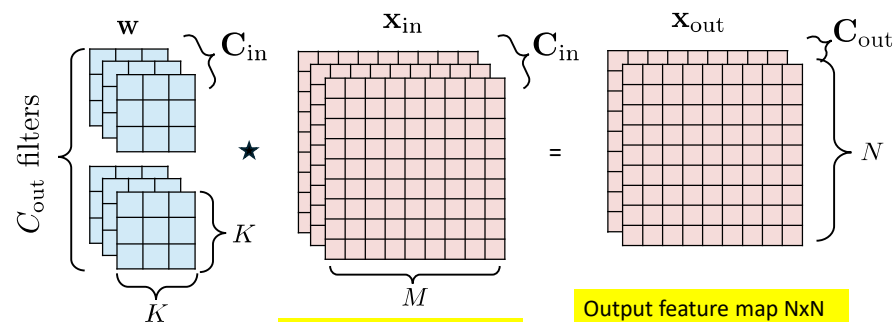
Channels in 2D Convolutions



Kernel size, stride, dilation all work as you would expect

32

General Convolutional Layer Form: Multi-Input, Multi-Output

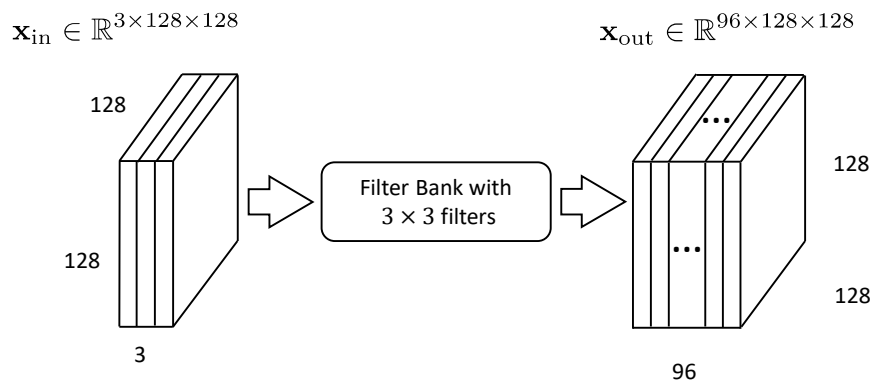


$$\mathbf{x}_{out}[c_2, :, :] = \sum_{c_1=1}^{C_{in}} \mathbf{w}[c_1, c_2, :, :] \star \mathbf{x}_{in}[c_1, :, :] + b[c_2]$$

Input channel Output channel

36

Multiple channels: Example

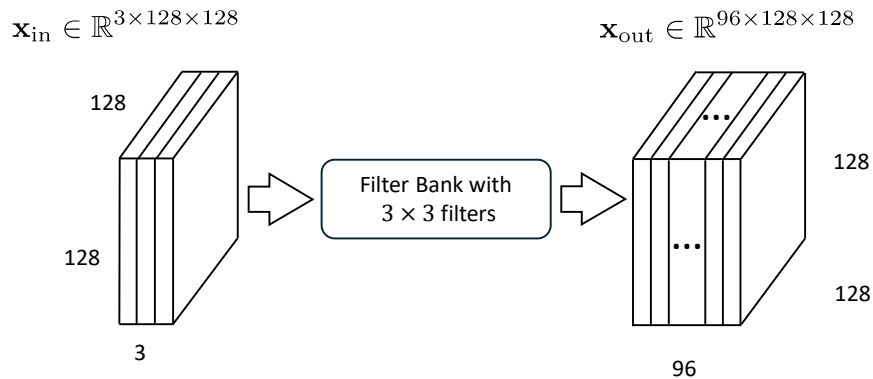


How many parameters does each filter have?

- (a) 9 (b) 27 (c) 96 (d) 864

37

Multiple channels: Example



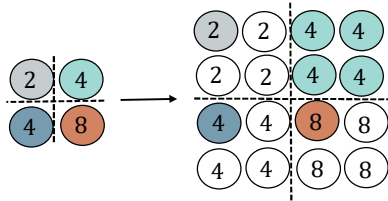
How many filters are in the bank?

- (a) 3 (b) 27 (c) 96 (d) can't say

38

Upsampling

a)

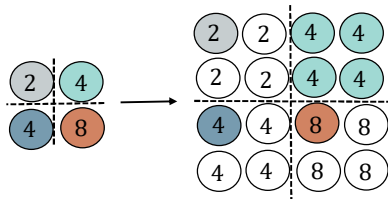


Duplicate

39

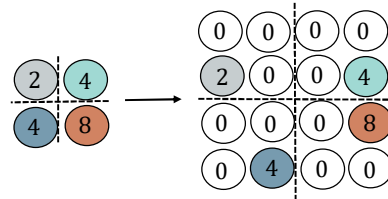
Upsampling

a)



Duplicate

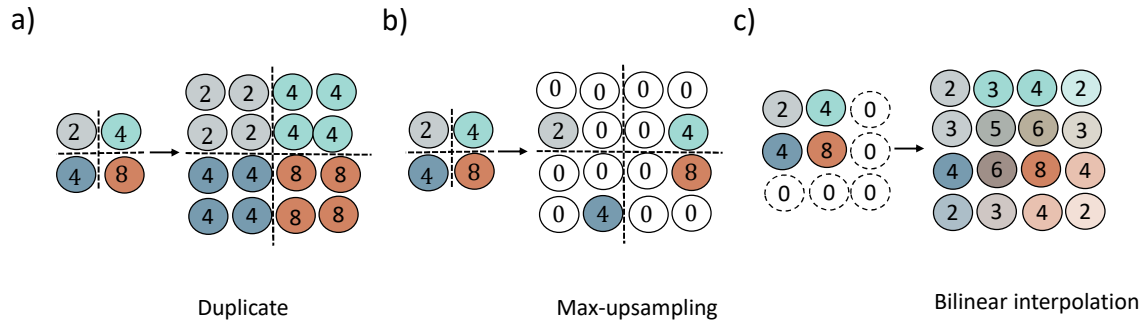
b)



Max-upsampling

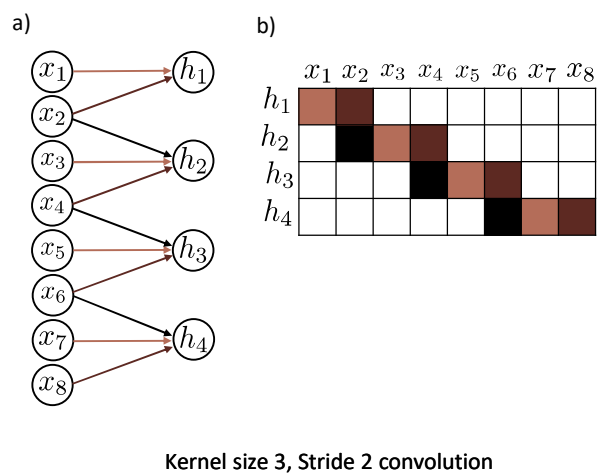
40

Upsampling



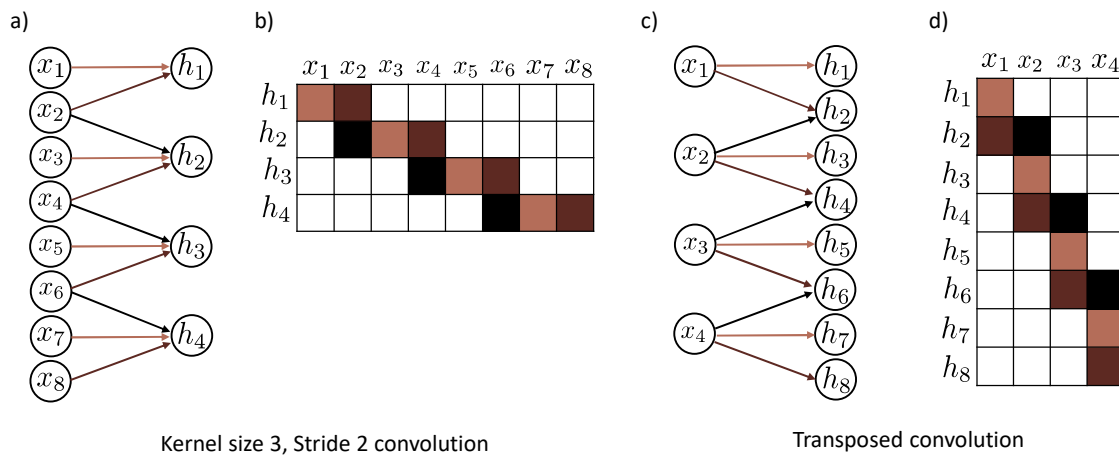
41

Transposed Convolutions



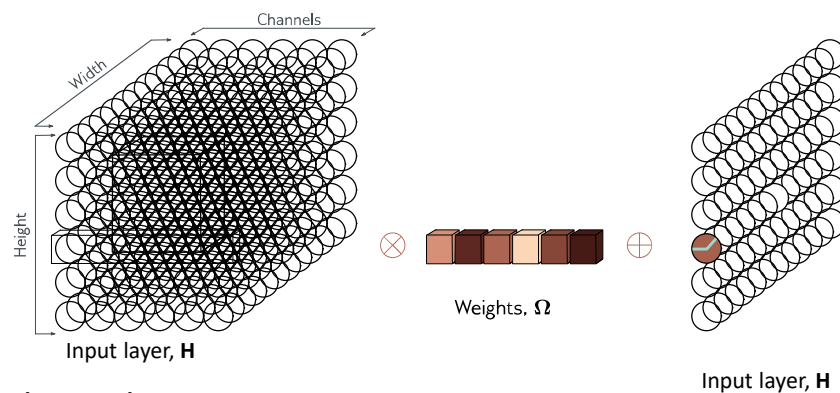
42

Transposed Convolutions



43

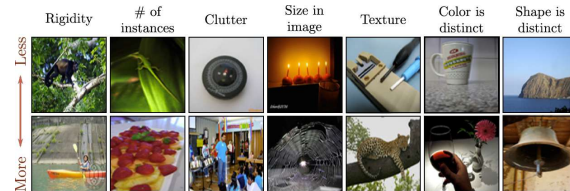
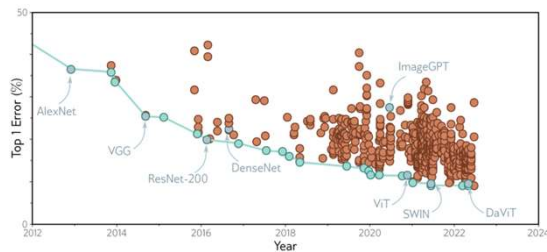
1 × 1 Convolutions



- Mixes channels
- Helps change number of channels
- Equivalent to running same fully connected network at each position

44

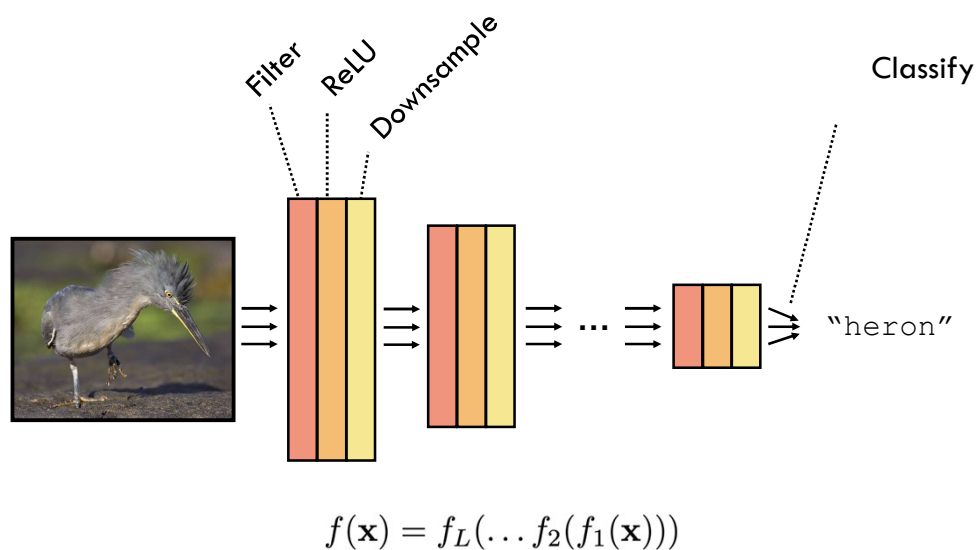
ImageNet



- Within a couple of years of Alexnet, matched human performance
 - 224 x 224 images
 - 1,281,167 training images, 50,000 validation images, and 100,000 test images
 - 1000 classes

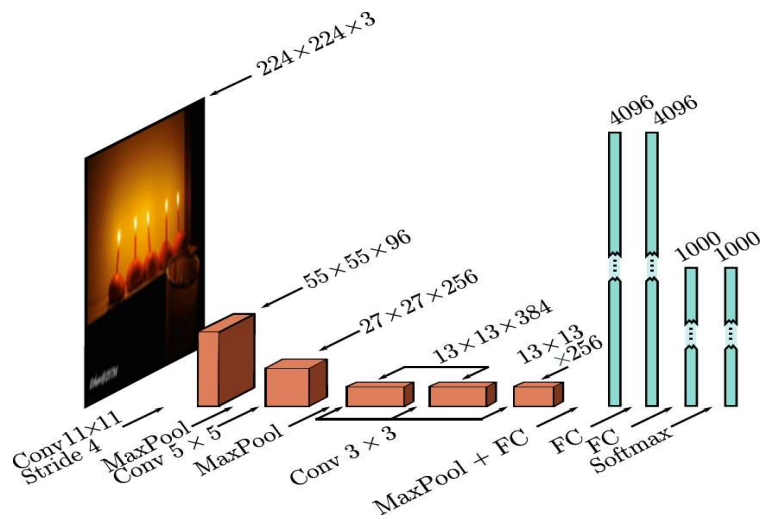
45

Computation in a neural net



46

AlexNet (2012)



- Almost all the 60 million parameters are in fully connected layers
- 16.4% top-5 error rate and a 38.1% top-1 error rate